Introduction to Hypothesis Testing

Given a coin – this coin is fair.

→ Test it using data

Hypothesis Testing is a statistical method used to make decisions about a population based on sample data.

Core Idea

1. Start with an assumption
2. Collect data
3. Decide whether accept or reject our assumption

1 Null Hypothesis (H_0)

Default assumption (status quo)

Assumption → nothing unusual
no change
no effect
no difference

Fair coin, fair dice

$=, >, <$

Ex A Teacher claims that the average marks of students is 70

Null Hypothesis $(H_0) = \mu = 70$
Alternative Hypothesis (H_1 or H_a) $\mu \neq 70$

2 Alternative Hypothesis

What you want to prove

"Something interesting is happening"

(H_1) Coin is biased $p \neq 0.5$

(H_1) Scores improved $\mu > 70$

Ha is a statistical statement that contradicts the null hypothesis and represent the effect, difference or relationship in the population parameter.

It is accepted when sufficient evidence to the reject the null hypothesis (H_0)

(H_0) $\mu = 70$

$\mu \neq 70 \rightarrow (H_1)$ true

Types of Alternative Hypothesis

1. Two-Tailed Alternative Hypothesis

Is it different from what we claimed?

just check difference → whether increase or decrease

$$H_1 : \mu \neq \mu_0$$

Ex An institute claims that the average scores of students is 70.

⇒ Students feel → Paper was tough
Or evaluation changed

⇒ So we want to test whether the avg score is different from 70, in any direction.

Hypo : $H_0 : \mu = 70$

$H_1 : \mu \neq 70$

Interpret: sample mean $> \mu$, we reject H_0

sample mean $< \mu$, we reject H_0

iff sample mean $\simeq \mu$, we fail to reject H_0

2 Right Tailed Alternative Hypothesis

We are checking only for an increase

$\Rightarrow$ Has it improved / increased?

$$H_1 : \mu > \mu_0$$

Ex Using a new method of teaching and claim that it improves the student's performance.

$\mu = 70$

After using the new method, we collect data

$\Rightarrow$ Did the scores increase

$H_0 = \mu = 70$ (no improvement)

$H_1 = \mu > 70$ (scores improvement)

Interpret: Sample mean > 70, reject H_0 accept H_1

If sample <= μ , we can't reject H_0

Even if scores are decreasing we can't reject H_0

$\Rightarrow$ As our test only checks improvement

Ex A company gives bonuses only if:

Average sales per employee exceeds ₹ 1 L

We test: Did the employees cross this threshold?

$H_0 : \mu = 1,00,000$ (No bonus)

$H_1 : \mu > 1,00,000$ (Provide bonus)

Left - Tailed Hypothesis Testing

Checking for a decrease

"Has it reduced?"

$H_1 : \mu < \mu_0$

Ex Packaged food company claim the each packet contains 200 g of product

Customers suspect that the company is underfilling the packets to save cost.

$H_0 : \mu = 200$ (no cheating)

$H_1 : \mu < 200$ (underfilling)

Interpret: $\mu < 200$ reject H_0

if $\mu >= 200$, we don't care
can't reject H_0

Type

2-Tailed — Is it different

Right Tailed — Has it increased?

Left Tailed — Has it decreased?

Ex "A new diet claims to reduce weight"
Which test should we use?

$\mu = \underline{\underline{80}}$ > Left - Tailed

Type I Error | Type II Error

Type I Error

error of rejecting a true null hypothesis

Its probability is denoted as α (significance level)

Eg Medicine Testing

H_0 : Medicine has no effect
H_1 : Medicine works

Type I Error : We conclude the medicine works, but in reality, it doesn't

$\Rightarrow$ False Positive

Prob (Type I Error) : α

Type II Error:

We fail to reject the null hypothesis even though it is false.

Error of failing to reject a false null hypothesis.

Prob(Type II Error): β

Eg Medicine Testing

H_0: Medicine has no effect

H_1: Medicine works

Type II Error:

We conclude the medicine doesn't work, in reality, it actually works

$\Rightarrow$ False Negative

Reality	Decision	Error Type
H_0 is true	Reject H_0	Type I
H_0 is false	Fail to Reject H_0	Type II

Significance level

Prob (Type I Error) = α

Pre-specified threshold that defines that max allowable prob of incorrectly rejecting a true null hypothesis

H_0 is true
$\Rightarrow$ Is the data too unusual to belive this assumption?

α: How unusual is too unusual?
Cutoff

Imagine all possible outcomes when H_0 is true:
Most outcomes - common
Some outcomes $\rightarrow$ rare

α marks the extreme region

$\alpha = 0.05$

$\Rightarrow$ Only outcomes that happen less than 5% of the time are considered to be rare.

$\Rightarrow$ If H_0 is actually true, we will wrongly reject it 5% of the time.

Case 1 $\alpha = 0.01$ (Very Strict)

Harder to reject H_0

$\Rightarrow$ Fewer false alarms

$\Rightarrow$ But may miss real effects

Case 2 $\alpha = 0.10$ (Less Strict)

Moderately rare events cause rejection

Easier of reject the H_0

$\Rightarrow$ More false alarms

$\Rightarrow$ But detects more real effects

$\alpha = 0.05$ Balance

$\alpha \rightarrow$ How sensitive the system is?